

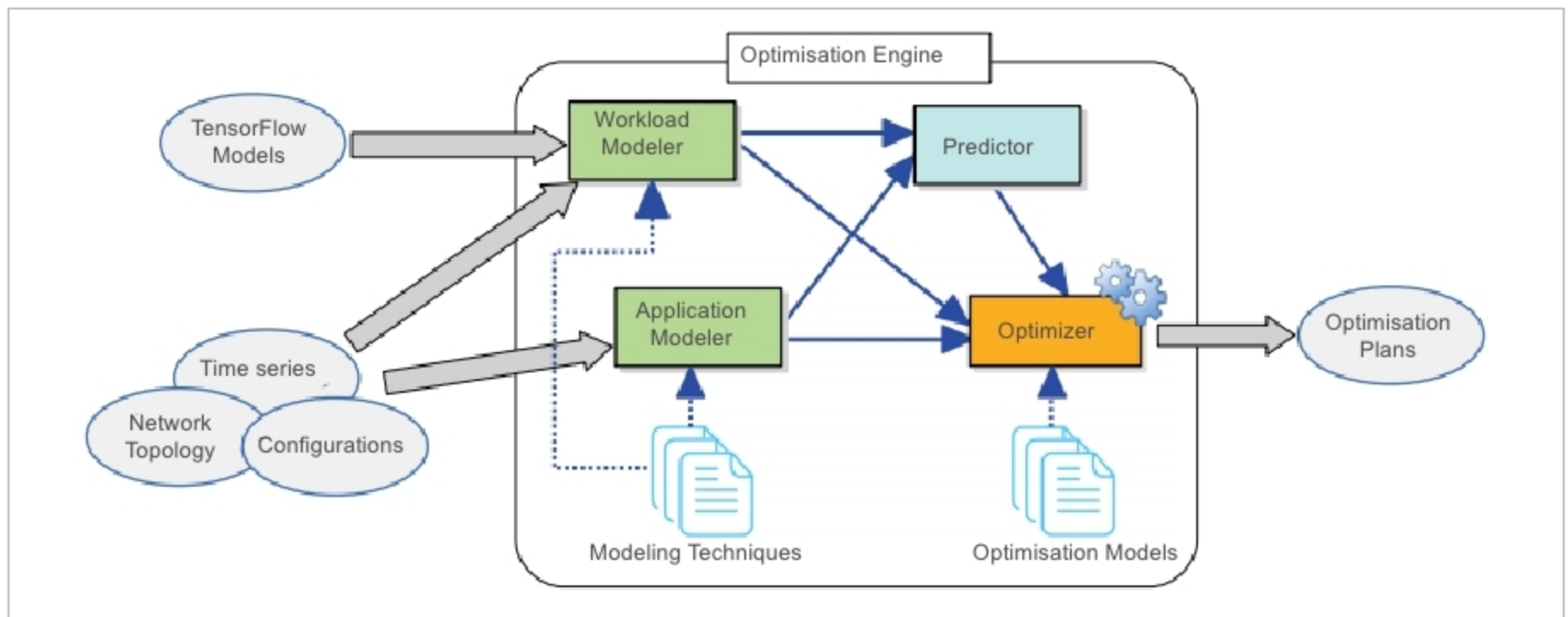


Figure 2: Cloud edge provisioning reference architecture [Reference: Machine Learning Methods for Reliable Resource Provisioning in Edge-Cloud Computing: A Survey, <https://dl.acm.org/doi/10.1145/3341145>, Fig. 4, P 9]

and IT are merged into the radio access network (RAN) in the proximity of the end user.

Application architecture is another important aspect.

The emergence of data intensive 5G mobile telephony mandates a reformed look at the system architecture. As of today, the typical cloud architecture consists of a set of services/components deployed across nodes. The shift of services from the traditional cloud deployments towards edge makes it possible for the application to be less network hungry and quicker in response. Hence, we see the proliferation of microservices beyond the traditional multi-tier service-oriented architectures.

Last but not the least, another important aspect is application migration. Since services are hosted in containers, VMs, etc, a migration here refers to the VM/container's migration while saving the present execution state to make it seamless.

A reference architecture

Figure 2 gives a helicopter view of a sample reference architecture.

Here we need two types of modelling. The first one is to understand the workload pattern and the second is

to understand the application pattern. The predictor predicts the workload and application behaviour. The optimiser yields the optimisation plan to place the VM/container working on the prediction values.

The workload modeller takes two sorts of input: raw data in terms of current metrics, network topology and the configurations; and models generated by the third party modelling tools. The predictor depends not only on the workload modeller output, but also on the application modeller's output to manage the workload propagation or workload distribution throughout the components of the application.

The optimiser also evaluates multiple models to fine tune the

optimisation plans for the controlled placement of the components across available resources under the SDI abstraction.

Techniques

As we now have a glimpse of the architecture, let us now look into some implementation details.

Workload modelling, analysis, prediction: Figure 3 depicts one of the many ways to synthesise a workload fingerprint for systems under deployment.

Placement optimisation: Deploying application components into bare metal servers, VMs, containers, etc, raises different issues related to replicas or content placements similar to those in CDN. The problem has its own set of constraints primarily coming from various key performance

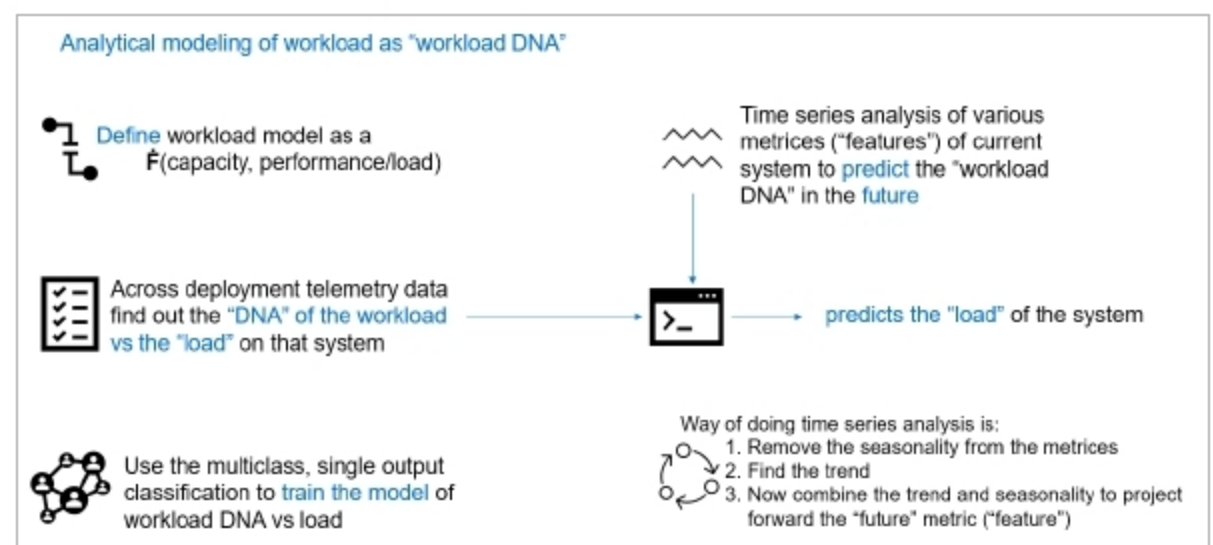


Figure 3: Workload DNA analytical modelling